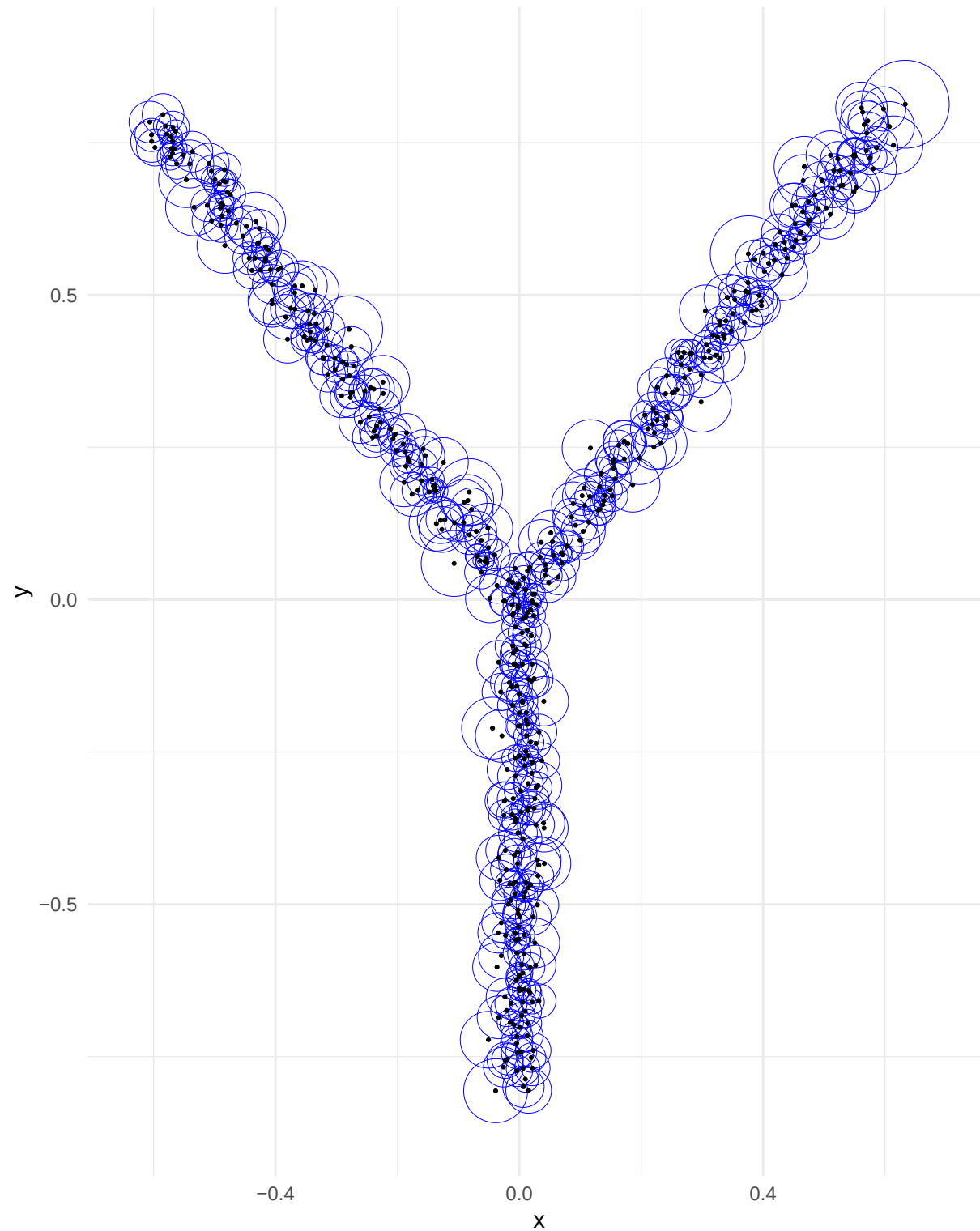
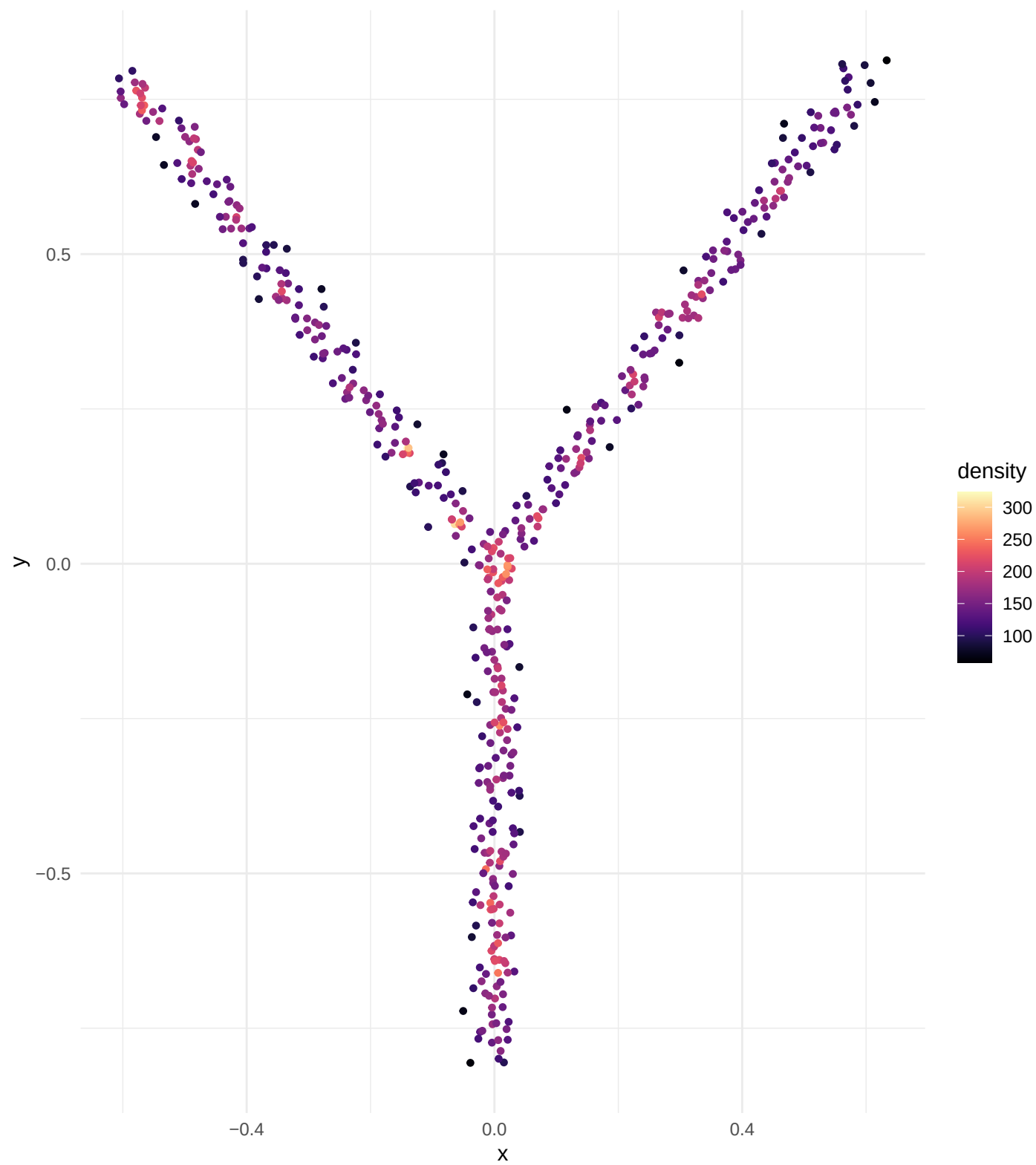


All Adaptive Spheres

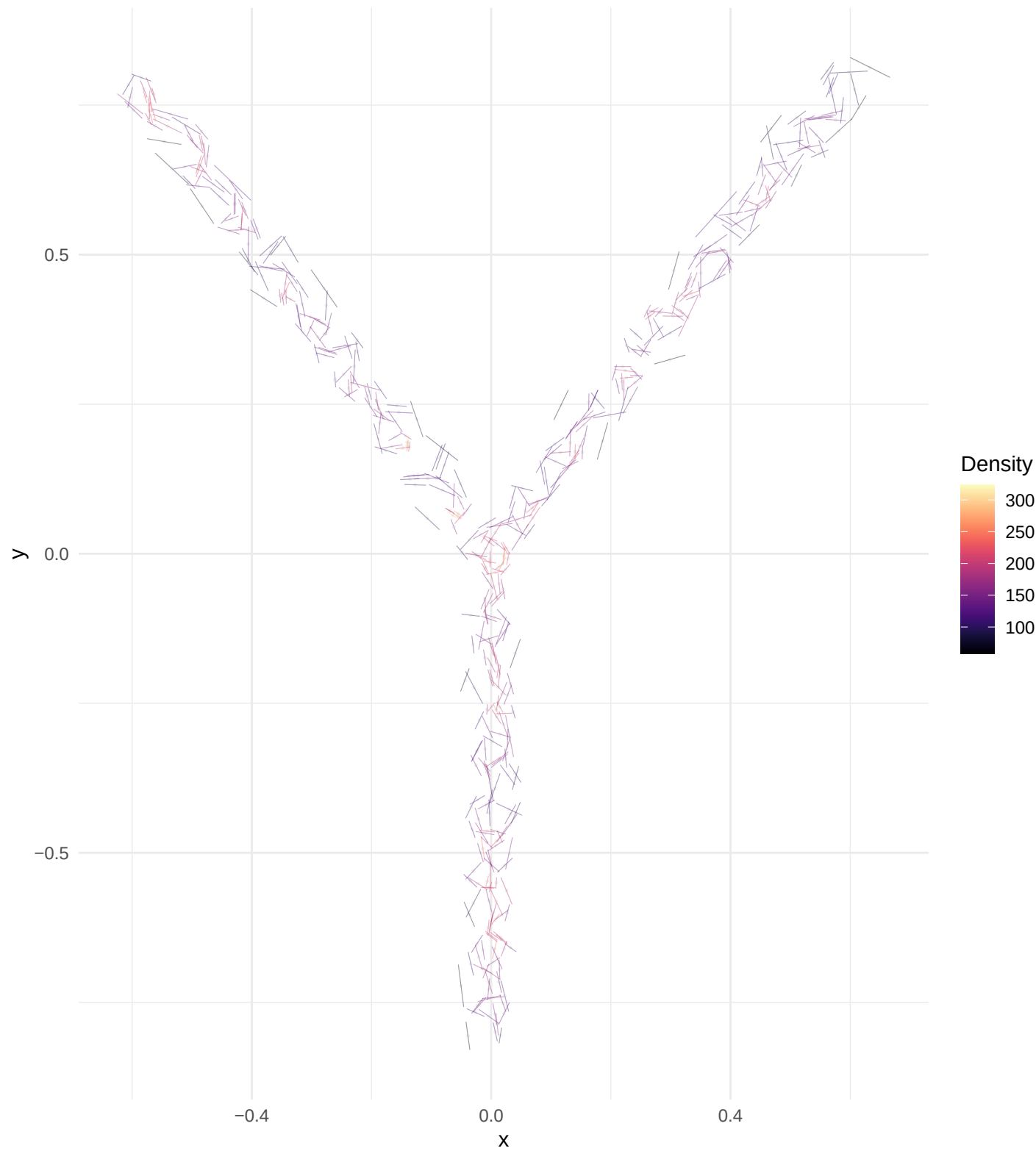
Visualizing local neighborhood size (radius) for every point.



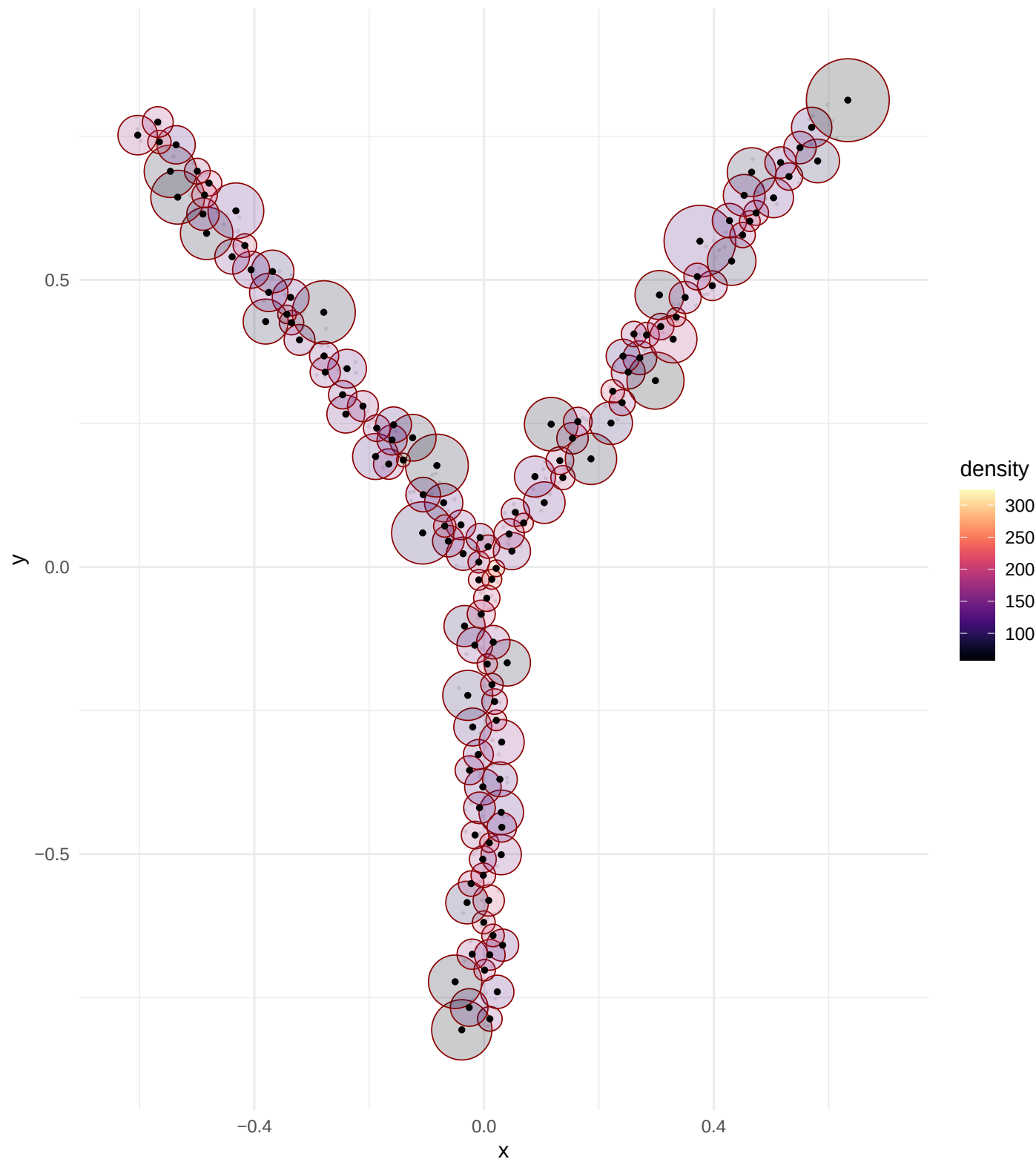
Point Density Distribution



Global Tangent Field (PC1)

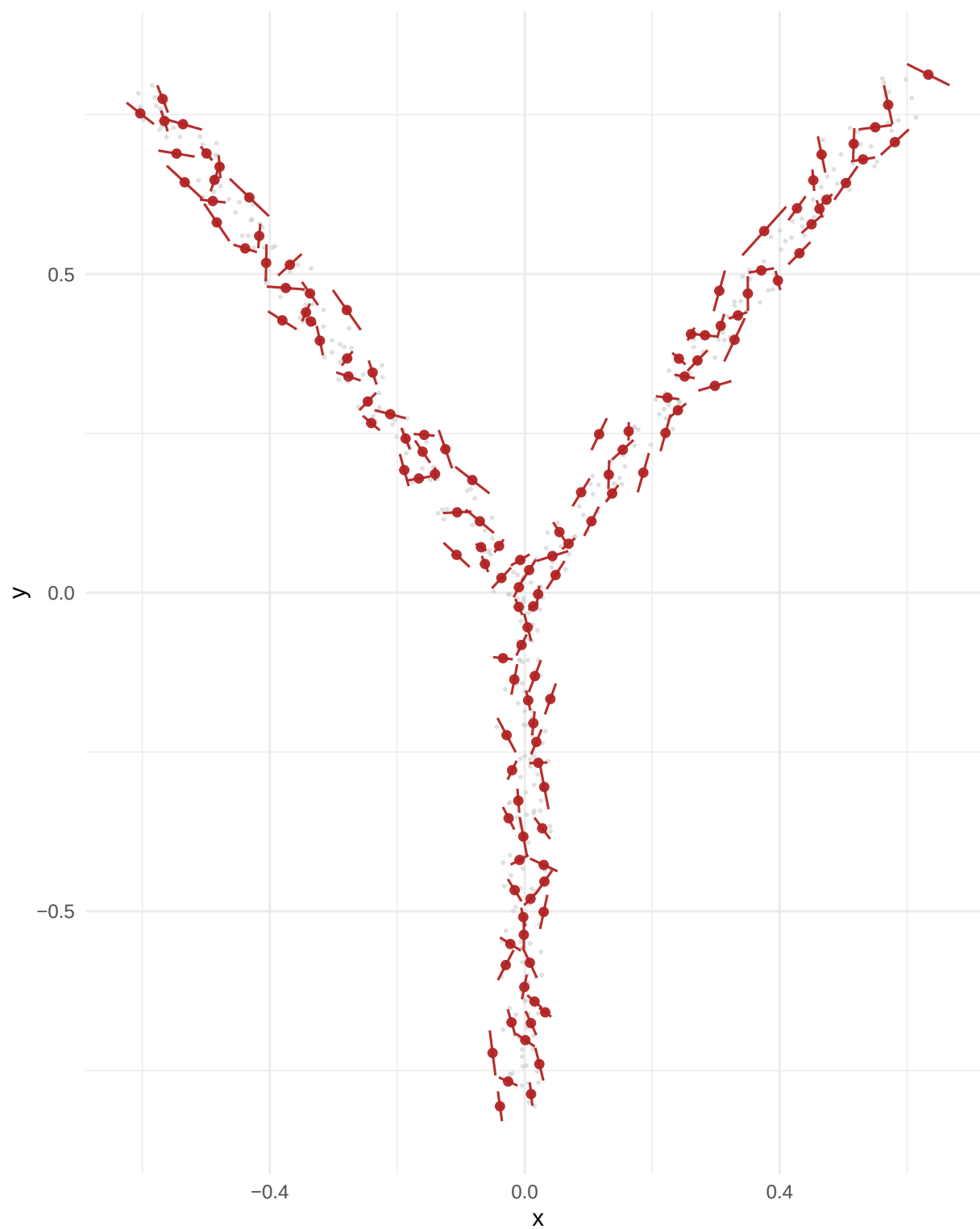


Atlas Selection (Anchors)



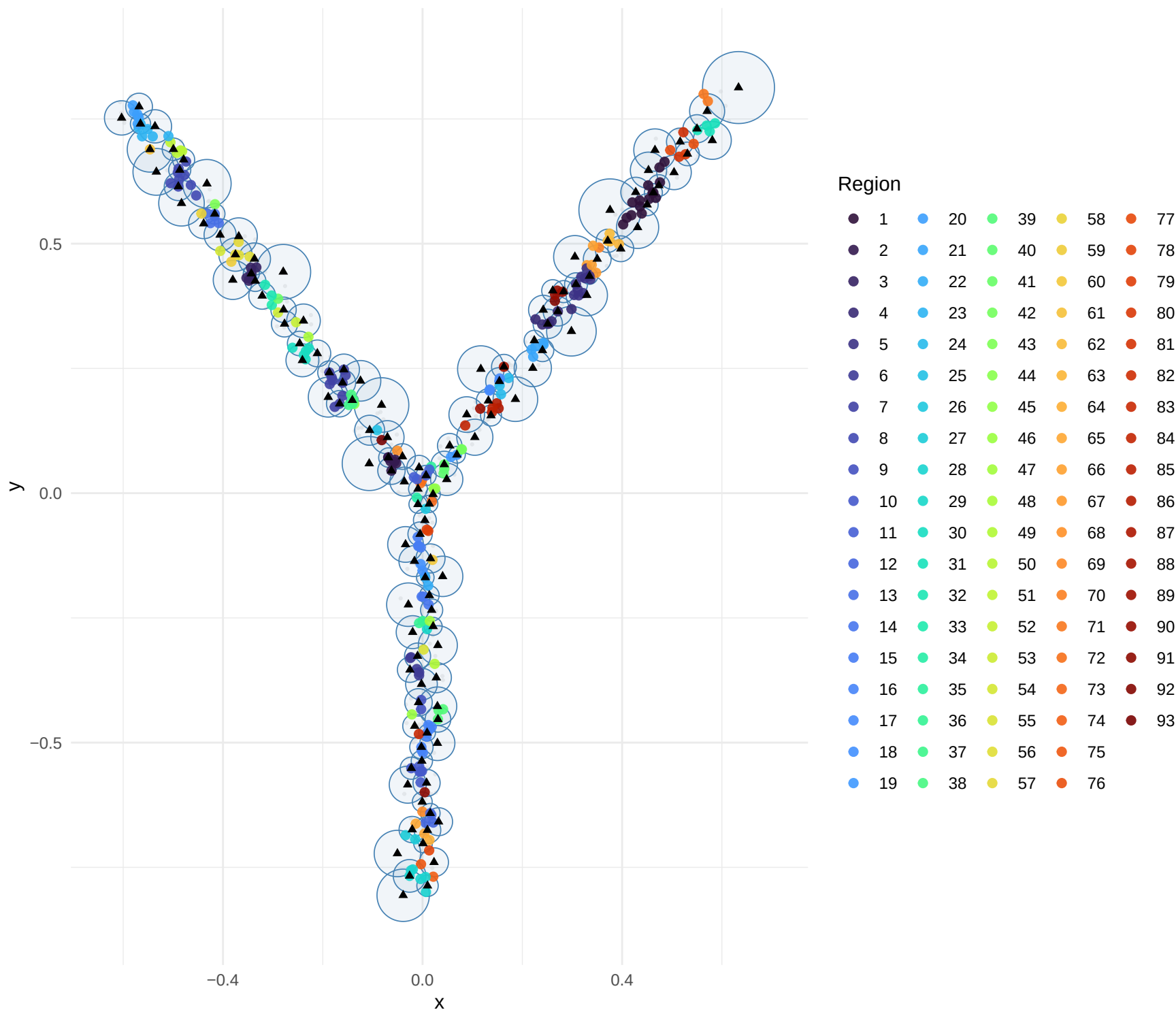
Anchor SVD Segments

Same anchors as plot 4: spheres first, then their SVD (PC1) in red



Intersection Regions (2+ anchors)

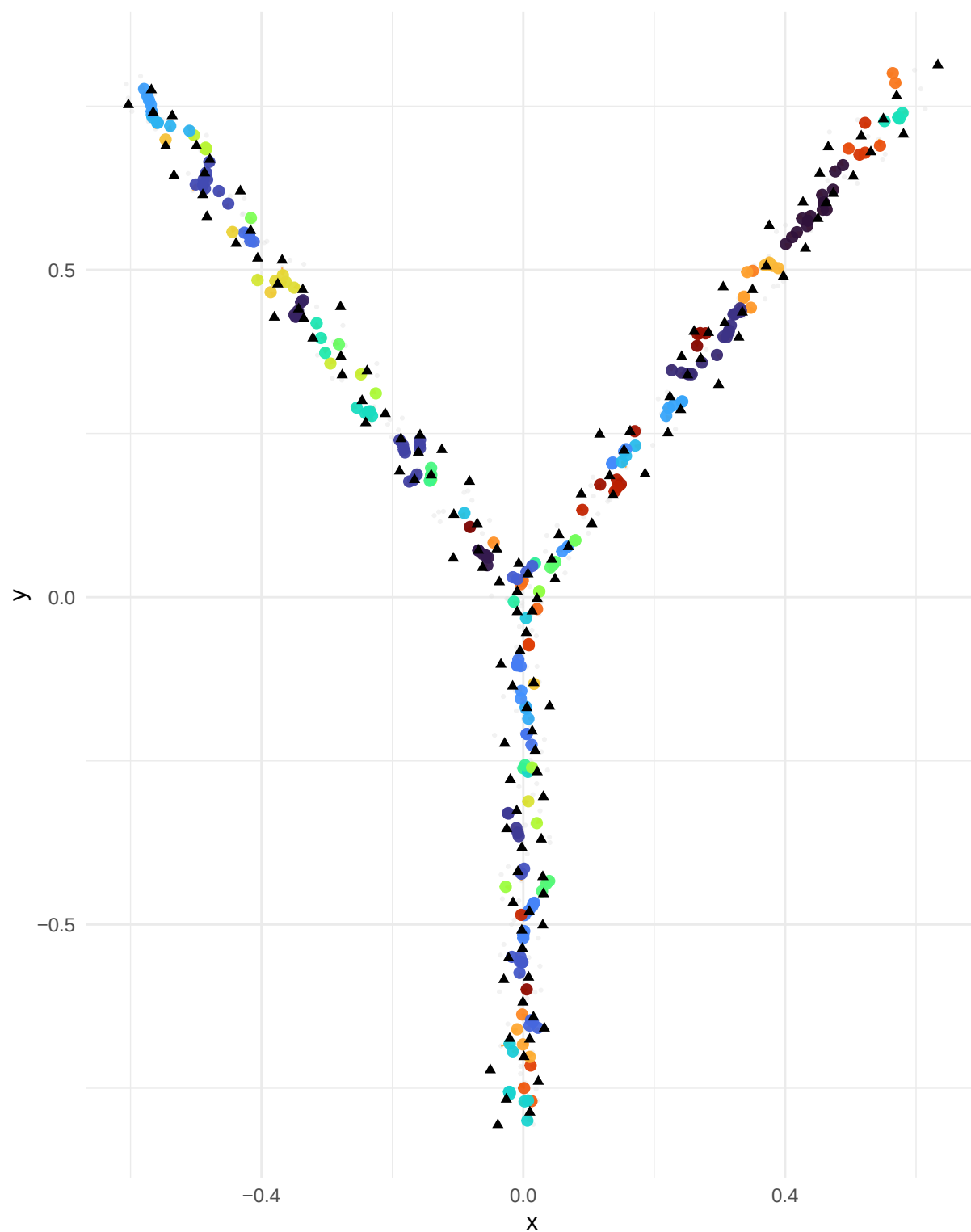
Anchor spheres in blue. Colored points = overlap zones. Triangles = anchor centers.



Parameters: $k_{\min} = 5$, $g_{\text{threshold}} = 3.0$, $o_{\max} = 0.3$, $o_{\min} = 0.10$, $\text{stability} = 0.90$, $\text{scale_factor} = 2.5$, $\text{max_iter} = 50$, $\text{conv_tol} = 0.01$

Stitched Backbone (Iteration 1)

Orange path: stitching; Triangles: original anchors

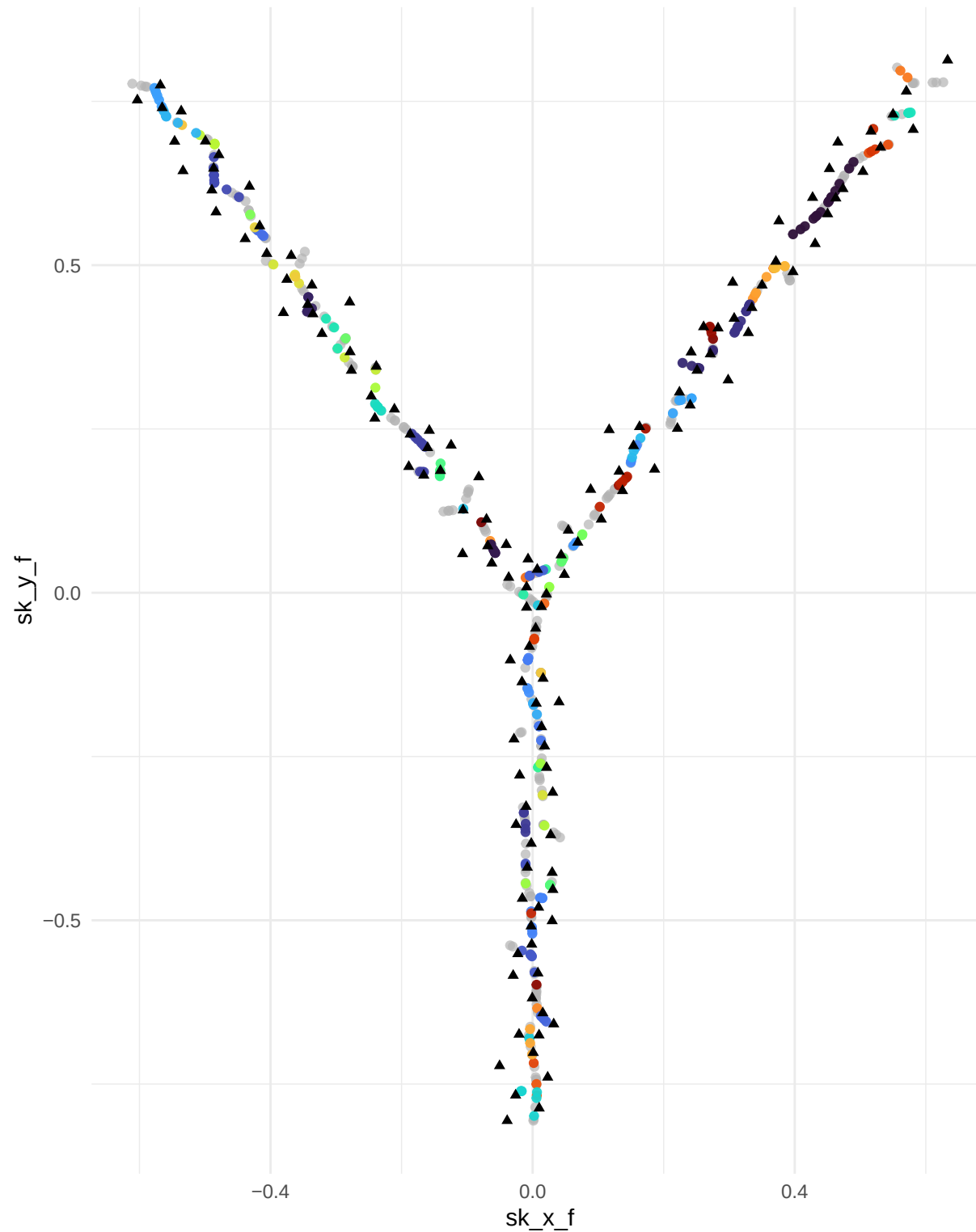


Region

1	20	39	58	77
2	21	40	59	78
3	22	41	60	79
4	23	42	61	80
5	24	43	62	81
6	25	44	63	82
7	26	45	64	83
8	27	46	65	84
9	28	47	66	85
10	29	48	67	86
11	30	49	68	87
12	31	50	69	88
13	32	51	70	89
14	33	52	71	90
15	34	53	72	91
16	35	54	73	92
17	36	55	74	93
18	37	56	75	
19	38	57	76	

Final Stitched Backbone (Converged)

Final stitched points



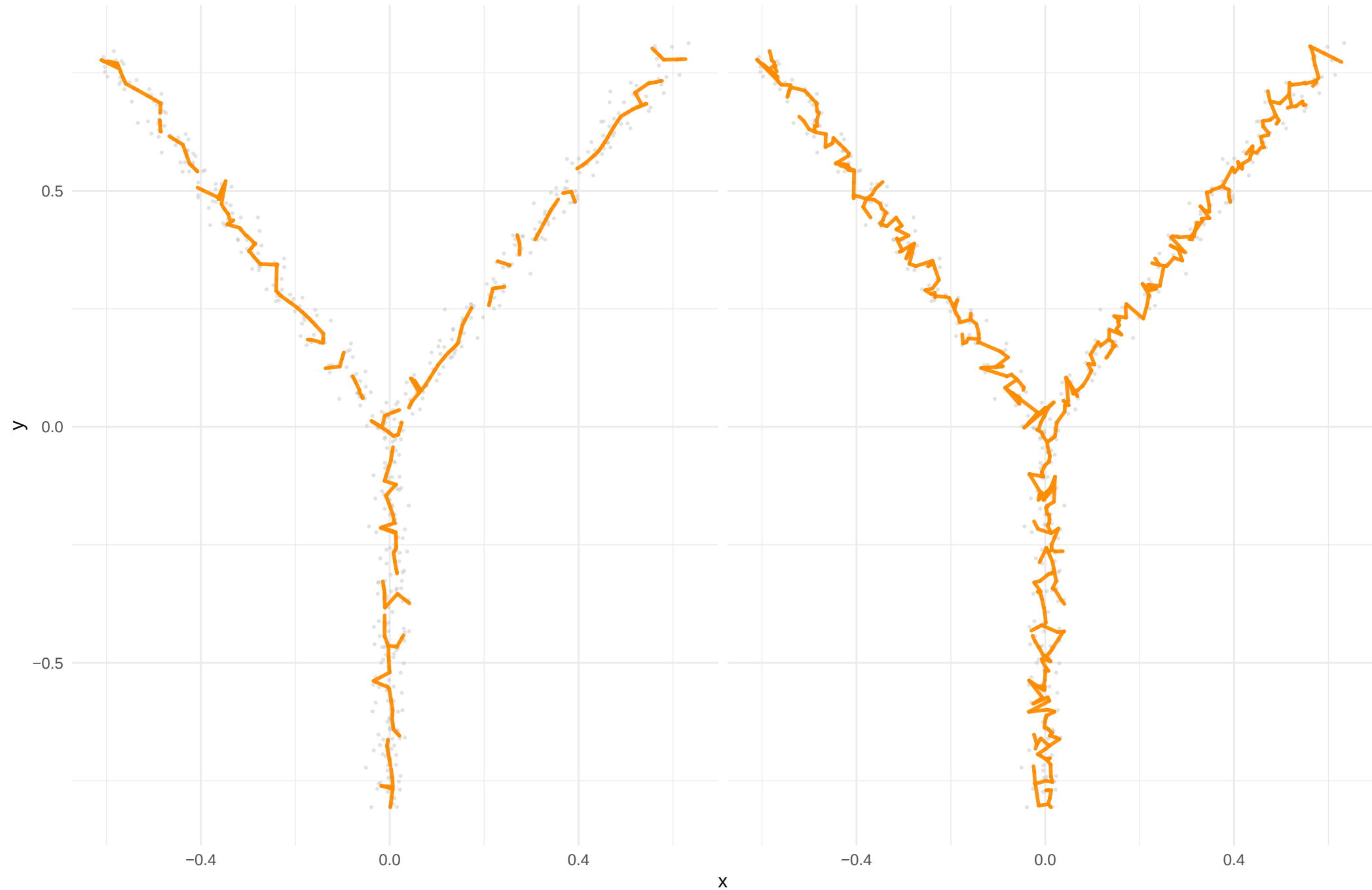
Region

1	20	39	58	77
2	21	40	59	78
3	22	41	60	79
4	23	42	61	80
5	24	43	62	81
6	25	44	63	82
7	26	45	64	83
8	27	46	65	84
9	28	47	66	85
10	29	48	67	86
11	30	49	68	87
12	31	50	69	88
13	32	51	70	89
14	33	52	71	90
15	34	53	72	91
16	35	54	73	92
17	36	55	74	93
18	37	56	75	
19	38	57	76	

Backbone Edges

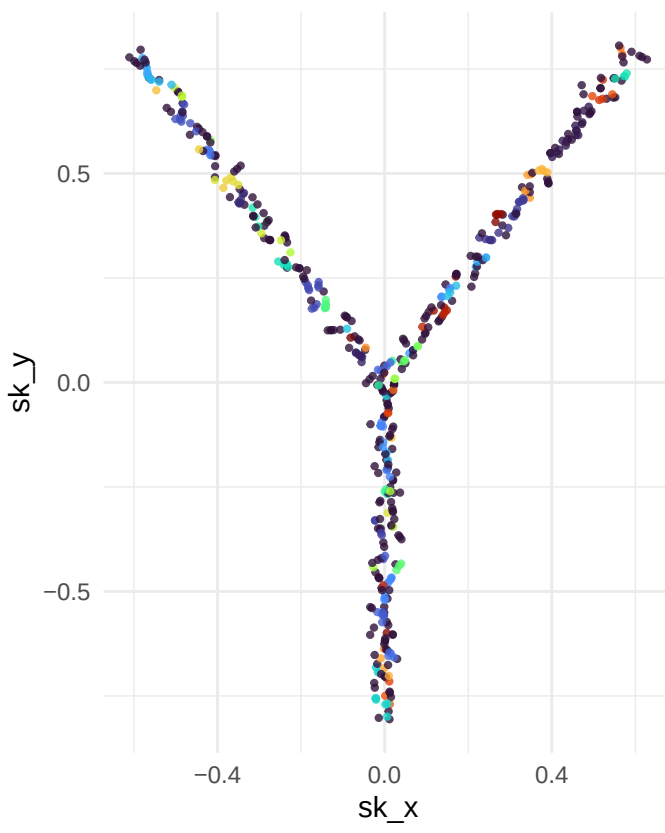
Final

Iteration 1

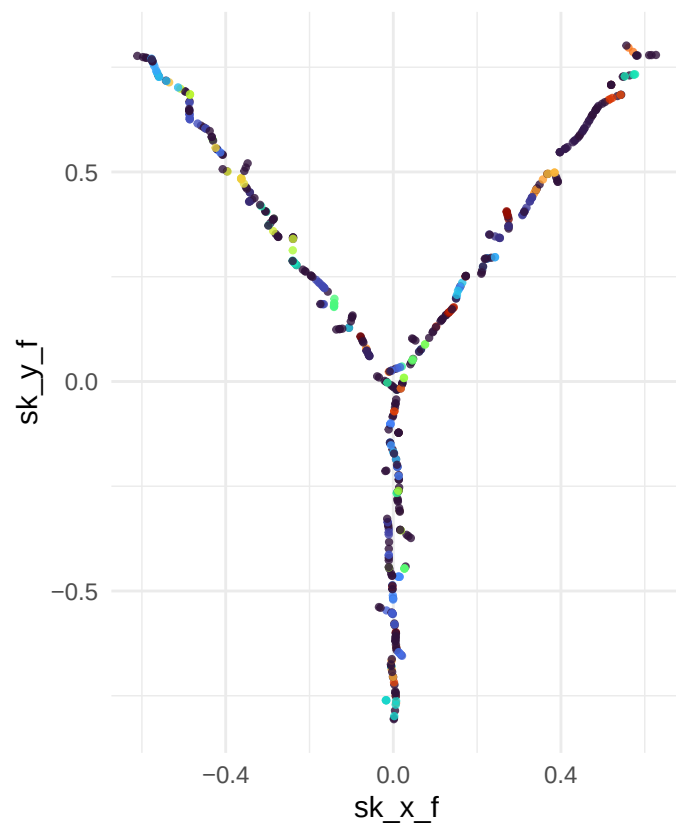


Convergence: iteration 1 vs final

Iteration 1

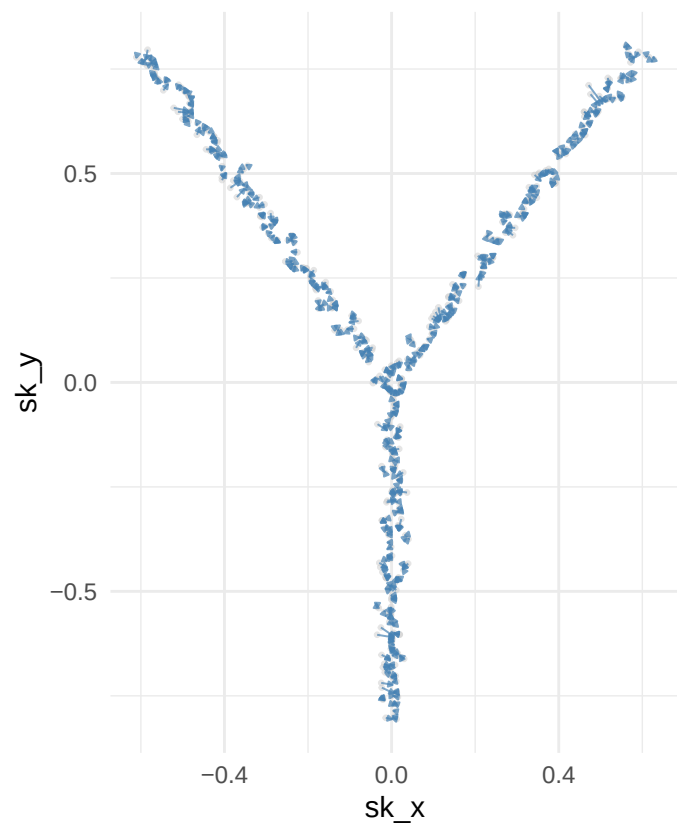


Converged



Displacement iter1 -> final

Points moved > 1e-6: 500 / 500



Parameters: k_min = 5, g_threshold = 3.0, o_max = 0.3, o_min = 0.10, stability = 0.90, scale_factor = 2.5, max_iter = 50, conv_tol = 0.01